NeST SE(3) decode tree on one real YCB-V image: the beam-MAP path through the joint tree

each level is a 64-way child = 8 rotations (shaded S^2 globes; filled HEALPix cells = density, hue = Hopf roll fibre) × 8 translation octants (octree cube on the MAP node)

gold = the beam-MAP root→leaf path (branch width ∝ joint probability); the bin+fibre refines rotation ×8 and translation ×8 each level

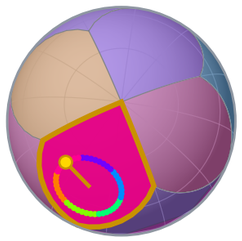


actual input
(obj 3: sugar box)

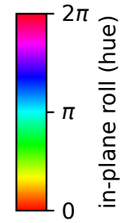
ViT-B/16 encoder

tokens condition every node

root $p_0(c_0 \mid x)$
72-way rotation



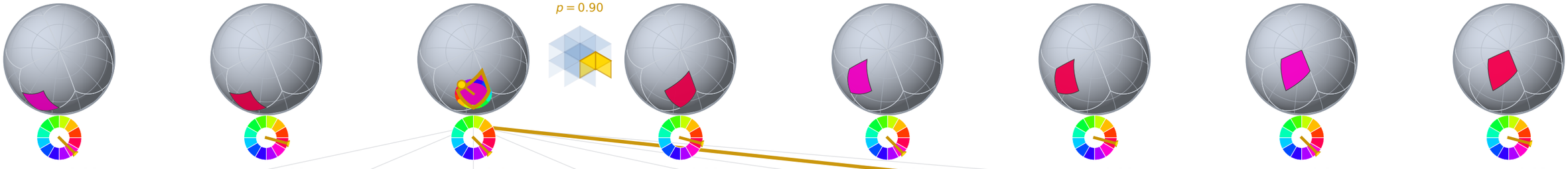




trans octant $(+\,-\,-)$
$p = 0.90$

$\ell=1$

64-way
8 × 8
nside 2

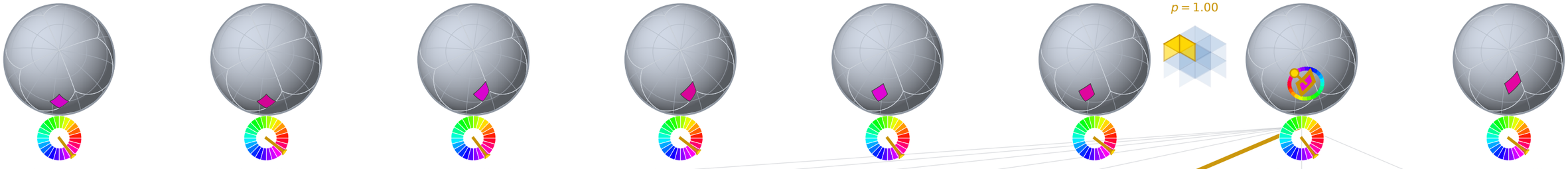


rot $p = 0.00$ | rot $p = 0.00$ | **rot** $p = 0.64$ | rot $p = 0.00$ | rot $p = 0.00$ | rot $p = 0.00$ | rot $p = 0.34$ | rot $p = 0.02$

trans octant $(-\,+\,+)$
$p = 1.00$

$\ell=2$

64-way
8 × 8
nside 4

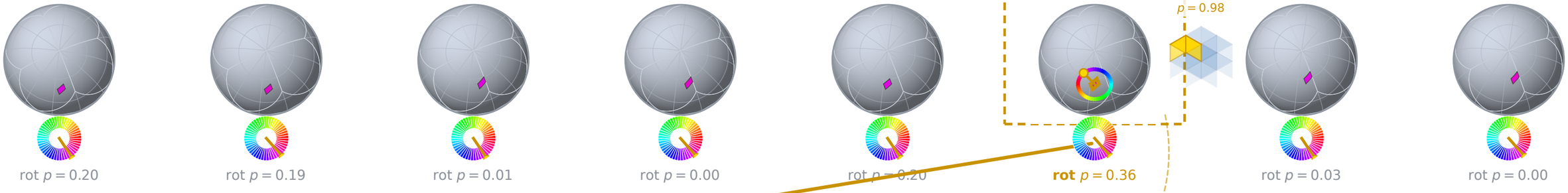


rot $p = 0.05$ | rot $p = 0.00$ | rot $p = 0.03$ | rot $p = 0.00$ | rot $p = 0.59$ | rot $p = 0.04$ | **rot** $p = 0.28$ | rot $p = 0.01$

trans octant $(-\,+\,+)$
$p = 0.98$

$\ell=3$

64-way
8 × 8
nside 8



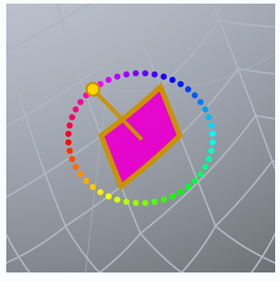
rot $p = 0.20$ | rot $p = 0.19$ | rot $p = 0.01$ | rot $p = 0.00$ | rot $p = 0.20$ | **rot** $p = 0.36$ | rot $p = 0.03$ | rot $p = 0.00$



ground truth — NeST MAP

leaf node =
output 6D pose → 3D box
scene 51/1629: rot 6.8°, **trans 8**

zoom: the $\ell=3$ MAP bin + fibre



nside 8
pix 222
roll 41/48
rot $p = 0.36$